COMPOSITIONS OF FLAT RING MAPS AND FLATNESS OF MODULES ALONG SCALAR RESTRICTION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted proof refers to two same-chapter lemmas by label; both are recorded as navigation stubs pointed at their \label lines, and the arguments consuming them stay inside the counted span.

Editorial note. The counted spans contain no \cite, so the card carries no bibliography.

Allowed standard prerequisites (editorial).

- The canonical isomorphism rewriting a twice-extended tensor product. Invoked in the counted proof at lines 8910-8914. Identifies the tensor products over the middle ring with the tensor products over the base ring.
- Flatness as preservation of exactness. Used twice in the counted proof (lines 8903-8909). Keeps the sequence exact after each of the two base changes.

Source reference. Original lemma 00CY: sources/stacks-algebra/pinned-original-source.tex, source lines 1996–1996. Navigation pointer for \ref{lemma-flip-tensor-product} used inside the counted span (line 8914); the argument that consumes it is retained in the span.

Source reference. Original lemma 00D2: sources/stacks-algebra/pinned-original-source.tex, source lines 2098–2098. Navigation pointer for \ref{lemma-tensor-with-bimodule} used inside the counted span (line 8913); the argument that consumes it is retained in the span.

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *A composition of (faithfully) flat ring maps is (faithfully) flat. If $R \to R'$ is (faithfully) flat, and M' is a (faithfully) flat R'-module, then M' is a (faithfully) flat R-module.*

Proof. The first statement of the lemma is a particular case of the second, so it is clearly enough to prove the latter. Let $R \to R'$ be a flat ring map, and M' a flat R'-module. We need to prove that M' is a flat R-module. Let $N_1 \to N_2 \to N_3$ be an exact complex of R-modules. Then, the complex $R' \otimes_R N_1 \to R' \otimes_R N_2 \to R' \otimes_R N_3$ is exact (since R' is flat as an R-module), and so the complex $M' \otimes_{R'} (R' \otimes_R N_1) \to M' \otimes_{R'} (R' \otimes_R N_2) \to M' \otimes_{R'} (R' \otimes_R N_3)$ is exact (since M' is a flat R'-module). Since $M' \otimes_{R'} (R' \otimes_R N) \cong (M' \otimes_{R'} R') \otimes_R N \cong M' \otimes_R N$ for any R-module N functorially (by Lemmas 00D2 and 00CY), this complex is isomorphic to the complex $M' \otimes_R N_1 \to M' \otimes_R N_2 \to M' \otimes_R N_3$, which is therefore also exact. This shows that M' is a flat R-module. Tracing this argument backwards, we can show that if $R \to R'$ is faithfully flat, and if M' is faithfully flat as an R'-module, then M' is faithfully flat as an R-module. □